

ITA0609 - Machine Learning

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1 .A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using Bayesian Belief Networks and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

Introduction

A smart healthcare monitoring system can use **wearable sensors** to continuously collect physiological parameters such as:

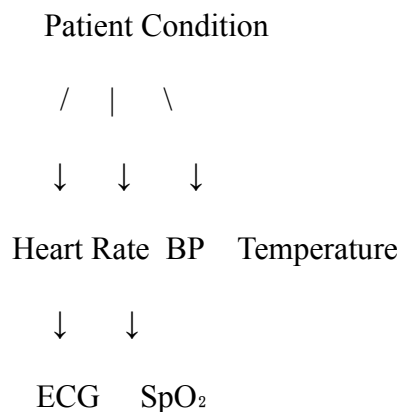
- Heart rate
- Blood pressure
- Body temperature
- Oxygen saturation (SpO₂)
- Respiratory rate
- ECG signals

Since sensor readings may contain noise and patients can have different normal ranges, **Bayesian Belief Networks (BBNs)** are useful for detecting abnormal conditions under uncertainty.

Bayesian Belief Network Model

A BBN represents physiological parameters as **nodes** and their conditional relationships as **directed edges**

For example :



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Health Status

The network calculates the probability of a patient's health condition based on observed sensor values.

For example:

$P(\text{Health} | \text{HR}, \text{BP}, \text{SpO}_2, \text{Temp})$

Using Bayes' theorem:

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

where:

- **H** = health condition
- **E** = observed sensor evidence

Conditional Dependencies

Physiological parameters are not completely independent.

For example:

- Abnormal heart rate can be related to abnormal ECG.
- Low SpO₂ can be associated with increased respiratory rate.
- Blood pressure can influence heart rate.
- Fever can affect heart rate and respiratory rate.

The BBN represents these relationships using **conditional probability tables (CPTs)**.

Heart Rate	SpO ₂	Probability of Abnormal Condition
Normal	Normal	0.05
High	Normal	0.30
Normal	Low	0.65
High	Low	0.95

Thus, multiple abnormal observations increase the probability of a serious condition.

Probability Learning

The probability values in the network can be learned from historical patient data.

Two common approaches are:

1. **Maximum Likelihood Estimation**

1. Estimates probabilities directly from observed data.

2. **Bayesian Parameter Learning**

1. Combines historical data with prior knowledge.
2. Useful when training data is limited.

The model can continuously update probabilities as new sensor observations arrive.

Handling Sensor Noise and Uncertainty

Wearable sensors can produce inaccurate readings because of:

- Movement
- Poor sensor contact
- Battery limitations
- Environmental interference
- Transmission errors

Instead of treating a reading as absolutely true, the Bayesian model assigns probabilities.

For example:

Observed $\text{SpO}_2 = 89\%$

Probability:

Normal reading = 0.15

Abnormal reading = 0.85

Therefore, one noisy measurement does not immediately generate an alarm. The system considers:

- Previous readings
- Other physiological parameters
- Sensor reliability
- Probability of the patient's condition

This reduces unnecessary alerts.

Anomaly Detection

The system continuously calculates:

$$P(\text{Normal} | \text{Evidence})$$

and

$$P(\text{Abnormal} | \text{Evidence})$$

If:

$$P(\text{Abnormal} | \text{Evidence}) > \text{Threshold}$$

the system generates an alert.

For example:

Heart Rate $\rightarrow$ High

SpO₂ $\rightarrow$ Low

Respiration $\rightarrow$ High

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Bayesian Network

↓

$$P(\text{Abnormal}) = 0.93$$

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HIGH-RISK ALERT

Challenges in Real-Time Deployment

1. **Sensor noise** – wearable devices may produce unreliable measurements.
2. **Real-time processing** – decisions must be generated quickly.
3. **Battery consumption** – continuous monitoring consumes battery.
4. **Network connectivity** – lost connections can cause missing data.
5. **Patient variability** – normal values differ between individuals.
6. **Privacy and security** – medical data must be protected.

7. **Scalability** – the system should support many patients simultaneously.
8. **False alarms** – excessive alerts can cause alarm fatigue.

Conclusion

A Bayesian Belief Network provides an effective approach for smart healthcare monitoring because it can model **relationships between physiological parameters and uncertainty in sensor measurements**. Probability learning allows the model to improve using historical patient data. With appropriate thresholds and real-time optimization, the system can detect anomalies early while reducing false alarms.

2.. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply Naïve Bayes classifiers, EM algorithm, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.

Introduction

A cybersecurity intrusion detection system (IDS) monitors network traffic and identifies whether network behavior is:

- **Normal**
- **Suspicious**
- **Malicious**

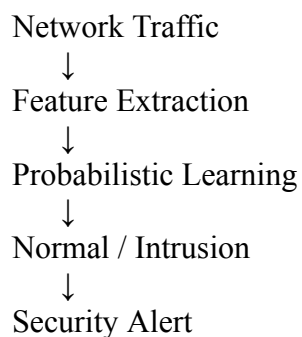
Probabilistic learning methods such as **Naïve Bayes, Expectation-Maximization (EM), and Bayesian learning** can be used to detect network intrusions even when data is incomplete or contains hidden patterns.

Network Behavior Model

Network traffic can be represented using features such as:

- Source IP
- Destination IP
- Protocol
- Port number
- Packet size
- Connection duration
- Number of packets
- Failed login attempts
- Bytes transferred

Example:



Naïve Bayes Classifier

Naïve Bayes calculates the probability that a network connection belongs to a particular class.

Using Bayes' theorem:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

where:

- **C** = class (Normal/Attack)
- **X** = network features

The Naïve Bayes assumption is that features are conditionally independent given the class.

Therefore:

$$P(X|C) = P(x_1|C)P(x_2|C)...P(x_n|C)$$

Example:

Packet Size = High
Failed Login = High
Connection Rate = High



Naïve Bayes



$P(\text{Attack}) = 0.94$



Intrusion

Advantages

- Fast classification
- Simple implementation
- Works well with large datasets
- Suitable for real-time intrusion detection

Limitation

The assumption that features are independent is often unrealistic in network traffic.

EM Algorithm

The **Expectation-Maximization (EM)** algorithm is useful when some network behavior is **hidden or unlabeled**.

It works in two steps:

1. Expectation Step

Estimate the probability that each data point belongs to a particular hidden class.

For example:

Network connection



70% Normal

30% Suspicious

2. Maximization Step

Update the model parameters based on these estimated probabilities.

The process is repeated:

Initial Model
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Expectation
↓
Maximization
↓
Updated Model
↓
Repeat until convergence

Bayesian Learning

Bayesian learning combines **prior knowledge** with new network observations.

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

where:

- **H** = intrusion hypothesis
- **D** = observed network data

As new traffic arrives, the system updates its belief.

For example:

Initial probability of attack = 20%

New suspicious traffic

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Bayesian update
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Probability of attack = 78%

Handling Hidden Patterns and Incomplete Data

Real-world network data may contain:

- Missing packet information
- Unknown attack types
- Encrypted traffic
- Unlabeled traffic
- Incomplete logs
- New attack patterns

EM algorithm

Can estimate missing or hidden information through iterative probability estimation.

Bayesian methods

Can incorporate prior knowledge and update beliefs when new evidence becomes available.

Naïve Bayes

Can still classify traffic using the available features, although missing attributes can reduce accuracy.

Reducing False Alarms

Traditional IDS systems can generate many false positives.

For example:

1000 suspicious connections

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100 detected as attacks

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30 are actual attacks

70 are false alarms

Performance Evaluation

The IDS can be evaluated using:

Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP+FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP+FN}$$

F1-score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Other important measures include:

- False Positive Rate
- False Negative Rate
- Detection latency
- Processing speed
- Resource consumption

For cybersecurity, reducing **false positives while maintaining high detection rates** is especially important.

Dynamic Environment

Network behavior continuously changes because of:

- New applications
- Changing user behavior
- Cloud services
- IoT devices
- New attack techniques
- Zero-day attacks

Bayesian learning can update probabilities when new data arrives, while EM can discover changing hidden patterns.

Therefore, the IDS can adapt instead of relying only on a static training dataset.

Challenges

1. **High network traffic volume**
2. **Real-time processing requirements**
3. **Changing attack patterns**
4. **Zero-day attacks**
5. **Incomplete and noisy data**
6. **Class imbalance**
7. **Encrypted network traffic**
8. **High false-positive costs**
9. **Adversarial attacks against ML models**
10. **Computational and memory requirements**

Conclusion

Naïve Bayes provides fast probabilistic classification, EM discovers hidden patterns and handles partially observed data, and Bayesian learning allows continuous updating of intrusion probabilities. Combining these techniques can produce an **adaptive intrusion detection system** with improved detection accuracy and fewer false alarms in dynamic network environments.

